2026 AI in Science Repository

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Preface

Science has always been a social and collective endeavor[1]. Furthermore, from the Manhattan Project to the modern AI race, many of the most consequential advances in human knowledge have come not from lone investigators – but from big teams of scientists[2]. In Fall of 2026, faculty, graduate students, and postdocs across multiple departments at the University of Florida leveraged these principles in a Fall 2026 seminar, titled Collective [Artificial] Intelligence in Science. Our goal? Leverage our own collective intelligence to rapidly make sense of one of the most pressing issues in contemporary research:

How is AI changing the nature of science, for better and for worse?

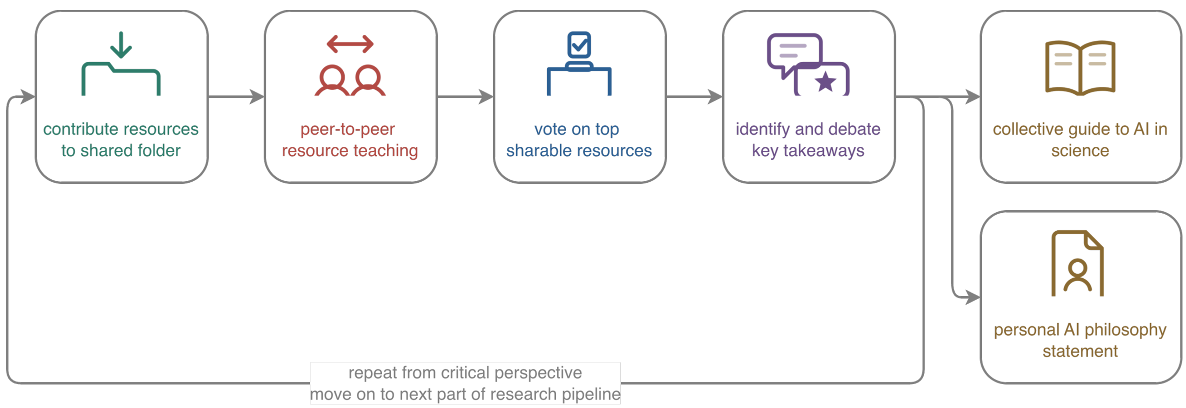

Figure 1: Structure of the University of Florida Fall 2026 seminar, titled Collective [Artificial] Intelligence in Science.

The seminar began with an introduction to big-team science – its evolution, competitive advantages, and relevance to the current moment[3,4]. From there, the course became a collective investigation, wherein we worked together to understand the newly disrupted nature of each stage of our research pipelines. This includes project management, literature review, idea development, data collection, analysis, writing, and dissemination[5]. To maintain balanced perspective, we alternated between deploying receptive vs. critical lens – identifying key resources, consensus statements, and areas of disagreement along the way. Throughout the semester, we organized our key insights into the resource you are viewing now: the 2027 Edition of the AI-in-Science Repository.

To develop this repository, each week followed a consistent structure (Figure 1). First, we would select a component of the research pipeline to examine (e.g., data analysis). Next, we would spend one week compiling resources for understanding the emerging role of AI in that part of the research pipeline. In class, we engaged in peer-to-peer teaching – all with the goal of identifying the top 3 resources to include in the repository. Last, we used a live onlined voting platform to solicit, debate, and vote upon key takeaways for the week (https://pol.is/). The result is the 2027 Edition of the AI-in-Science Repository: a summary of our top resources, takeaways, and disagreements.

Project Management

Project Management: Using AI

> *How is AI changing the way research teams plan, coordinate, and keep track of their work?*

Lens: receptive — what AI makes newly possible at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

Why it made the cut:

Who it is for:

2. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

3. *[Resource title]*

[Author(s), year]

What it is:

Why it made the cut:

Who it is for:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Project Management: Critiquing AI

What breaks when the coordination of a research team is delegated to AI?

Lens: critical — the failure modes, costs, and limits of AI at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

The critique it makes:

How strong is the evidence:

2. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

3. *[Resource title]*

[Author(s), year]

What it is:

The critique it makes:

How strong is the evidence:

Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
3. *[takeaway]*

Failure Modes to Watch For

- *[concrete way this goes wrong in practice, and the signal that it has]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Literature Review

Literature Review: Using AI

How is AI changing the way we find, triage, and synthesize prior work?

Lens: receptive — what AI makes newly possible at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

What it is:

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Key Takeaways

Statements that reached consensus in the Polis session.

1. *[takeaway]*
2. *[takeaway]*
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Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Literature Review: Critiquing AI

What does AI-mediated search hide, distort, or invent — and how would we notice?

Lens: critical — the failure modes, costs, and limits of AI at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

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How strong is the evidence:

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1. *[takeaway]*
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Failure Modes to Watch For

- *[concrete way this goes wrong in practice, and the signal that it has]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Idea Development

Idea Development: Using AI

How is AI changing the way research questions and hypotheses are generated?

Lens: receptive — what AI makes newly possible at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

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[Author(s), year — paper, tool, thread, talk, or documentation]

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Open Questions

- *[question the week raised but could not answer]*

Idea Development: Critiquing AI

What happens to originality and intellectual ownership when ideas are co-generated?

Lens: critical — the failure modes, costs, and limits of AI at this stage of the research pipeline.

Top Resources

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1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

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Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Data Collection

Data Collection: Using AI

> *How is AI changing the way we recruit participants, build instruments, and gather data?*

Lens: receptive — what AI makes newly possible at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

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[Author(s), year — paper, tool, thread, talk, or documentation]

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Open Questions

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Data Collection: Critiquing AI

What threats to data integrity, validity, and research ethics does AI introduce?

Lens: critical — the failure modes, costs, and limits of AI at this stage of the research pipeline.

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[Author(s), year — paper, tool, thread, talk, or documentation]

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Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Analysis

Analysis: Using AI

How is AI changing the way we write code, model data, and interpret results?

Lens: receptive — what AI makes newly possible at this stage of the research pipeline.

Top Resources

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[Author(s), year — paper, tool, thread, talk, or documentation]

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Points of Disagreement

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- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Analysis: Critiquing AI

Which analytic errors survive when the analysis is delegated?

Lens: critical — the failure modes, costs, and limits of AI at this stage of the research pipeline.

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Failure Modes to Watch For

- *[concrete way this goes wrong in practice, and the signal that it has]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Writing

Writing: Using AI

How is AI changing the drafting, revising, and labor of scientific writing?

Lens: receptive — what AI makes newly possible at this stage of the research pipeline.

Top Resources

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[Author(s), year — paper, tool, thread, talk, or documentation]

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Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Writing: Critiquing AI

What is lost when writing stops being thinking — and who counts as an author?

Lens: critical — the failure modes, costs, and limits of AI at this stage of the research pipeline.

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- *[concrete way this goes wrong in practice, and the signal that it has]*

Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Dissemination

Dissemination: Using AI

How is AI changing peer review, publishing, and how findings reach their audience?

Lens: receptive — what AI makes newly possible at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

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[Author(s), year]

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Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Dissemination: Critiquing AI

What does AI do to trust, gatekeeping, and the integrity of the scientific record?

Lens: critical — the failure modes, costs, and limits of AI at this stage of the research pipeline.

Top Resources

The three resources the seminar voted to carry forward.

1. *[Resource title]*

[Author(s), year — paper, tool, thread, talk, or documentation]

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Failure Modes to Watch For

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Points of Disagreement

Where the group did not converge, and what evidence would move it.

- *[position vs. counter-position — and what would settle it]*

Open Questions

- *[question the week raised but could not answer]*

Personal AI Philosophy Statements

Having examined the research pipeline together, where does each of us now stand?

The preceding chapters record where the seminar converged as a group. This chapter records where its members stand individually.

Alongside the shared repository, participants wrote personal AI philosophy statements: short, signed accounts of how each of us intends to use — and refuse to use — AI in our own research. Each statement is reproduced in full below and is free to share, quote, or adapt.

The statements

Nicholas A. Coles

[Role, department, institution]

[Statement.]

[Author]

[Role, department, institution]

[Statement.]

Where we agree

Commitments that recur across the statements.

1. *[point of agreement]*
2. *[point of agreement]*
3. *[point of agreement]*

Where we disagree

The substantive splits — not differences of emphasis, but positions that cannot both be right.

- *[position vs. counter-position — and what would settle it]*

Each statement speaks only for its author. Disagreement between them is the point, not an error to be corrected.

References

1. Merton, R. K. *On Social Structure and Science.* (University of Chicago Press, 1996).
2. Coles, N. A., Hamlin, J. K., Sullivan, L. L., Parker, T. H. & Altschul, D. Build up big-team science. *Nature* **601**, 505–507 (2022).
3. Wuchty, S., Jones, B. F. & Uzzi, B. The increasing dominance of teams in production of knowledge. *Science* **316**, 1036–1039 (2007).
4. Coles, N. A., Goes Braga Takayanagi, J. F., Grant, G. L. & Basnight-Brown, D. M. The rise, impact, and imbalances of big-team psychology. *Psychological Science* **37**, 287–297 (2026).
5. .